of authorization, not authentication. The client can typically not do anything about this other than ask the administrator for access.

3.6.1 Enforcing authentication

The most basic access control check is simply to require that all users are authenticated. This ensures that only genuine users of the API can gain access, while not enforcing any further requirements. We can enforce this with a simple filter that runs after authentication and verifies that a genuine subject has been recorded in the request attributes. Open the UserController.java file in your editor and add the following method, which can be used as a Spark `before()` filter to enforce that users are authenticated:

```
public void requireAuthentication(Request request,
    Response response) {
  if (request.attribute("subject") == null) {
    response.header("WWW-Authenticate",
        "Basic realm=\"/\", charset=\"UTF-8\"");
    halt(401);
  }
}
```

You can then open the Main.java file and require that all calls to the Spaces API are authenticated, by adding the following filter definition. Locate the line where you added the authentication filter earlier and add a filter to enforce